Karim Elgendi

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EDUCATION

George Mason University | Fairfax, VA

B.S. in Applied Computer Science, Concentration Software Engineer **May 2023 - December 2026**

Relevant Courses: Software Testing/Maintenance, Data Structures, Analysis of Algorithms, Operating Systems, Software Architecture

TECHNICAL SKILLS

Programming Languages: Java, C, SQL/PostgreSQL, HTML, CSS, JavaScript, Dart

Operating Systems: Windows, Linux (Ubuntu, Pop_OS!)

Technologies/Frameworks & Methodologies: GitHub, Git, Spring/Spring Boot, Agile, Scrum, Flutter, VS Code

PROJECTS

Personal Portfolio Project | HTML, CSS, JavaScript, Spring Boot, PostgreSQL **October 2025**

- Designing & developing a personal portfolio website through project-based learning using **HTML**, **CSS**, & **JavaScript**. Learning to utilize custom styling & responsive design principles.
- Utilized **GitHub** & **Git** through **VS Code** for version control, documenting the progress of development & challenges.
- Built a **Spring Boot REST API** with **CRUD endpoints**, using **Spring Security** to secure backend APIs that access the database and **Spring Data JPA** (**Hibernate**) for database persistence.

CPU Scheduler Project | C **May 2025**

- Developed and implemented core scheduling functions in **C** as a CPU Scheduler that manage process creations based on priority queues, & dealing with termination states within an emulated **Linux** Task Manager.
- Applied bitwise operations, linked list manipulation to streamline processes, & used processing techniques to handle multi-level feedback of queue scheduling & ensuring correct CPU allocation to prevent starvation.
- Gained hands-on experience in memory management, & error-handling practices to achieve functionality with zero memory leaks using **Valgrind**.

EXPERIENCE

Undergraduate Research Assistant | George Mason University **February 2026**

- Conducted targeted research to identify additional scientific GitHub repositories within a defined domain, expanding the pool of projects for comparative analysis and documentation review.
- Analyzed repository documentation to compare structure, onboarding processes, and maintenance practices, and to evaluate consistency across selected repositories.
- Assessed scientific GitHub project datasets for recency, completeness, and scalability by reviewing repository activity, documentation quality, and overall suitability for broader research use.
- Performed cross-repository comparisons to identify similarities, differences, and trends in project organization, documentation standards, and long-term maintainability.